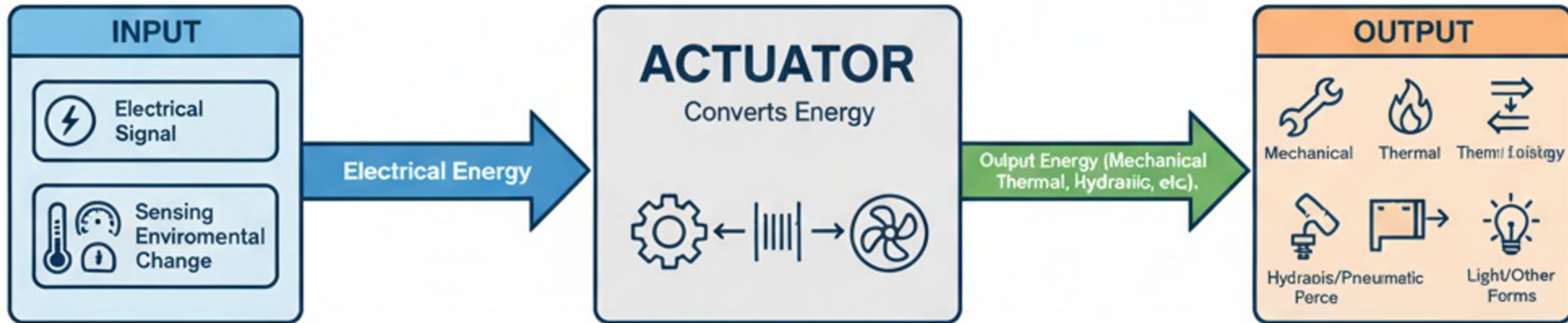


Actuators

Actuators

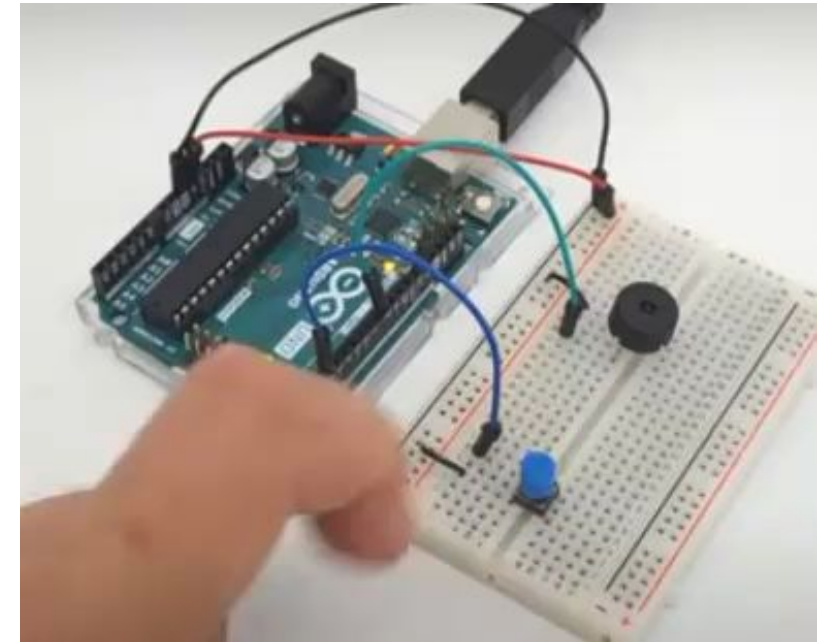
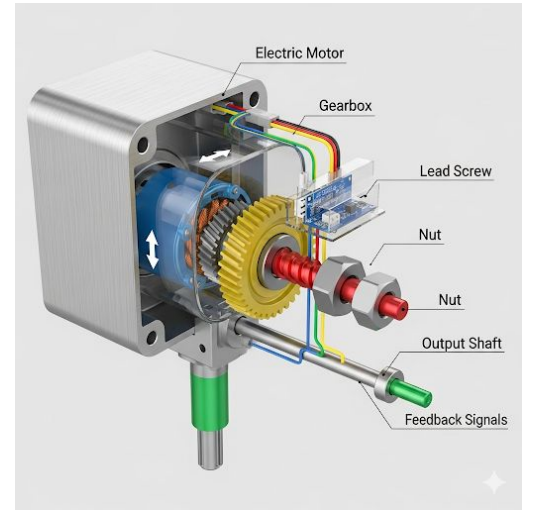
- An actuator is a device that converts the effective change in the environment sensed in the form of electrical energy into some other form of energy.



Based on their operation, these are classified into Electrical, Hydraulic, and Pneumatic actuators.

Electrical Actuator

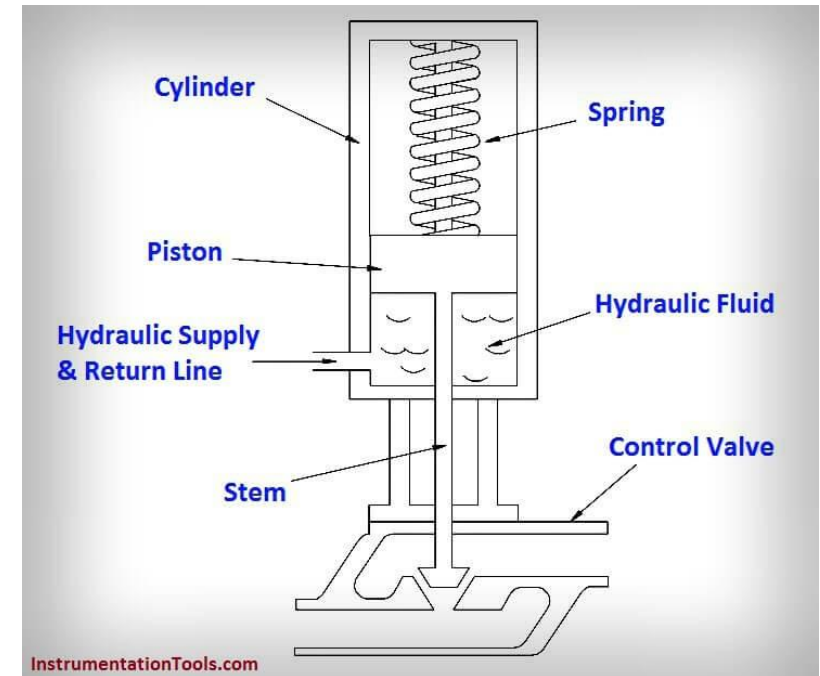
- Uses electrical energy



Hydraulic and Pneumatic Actuator

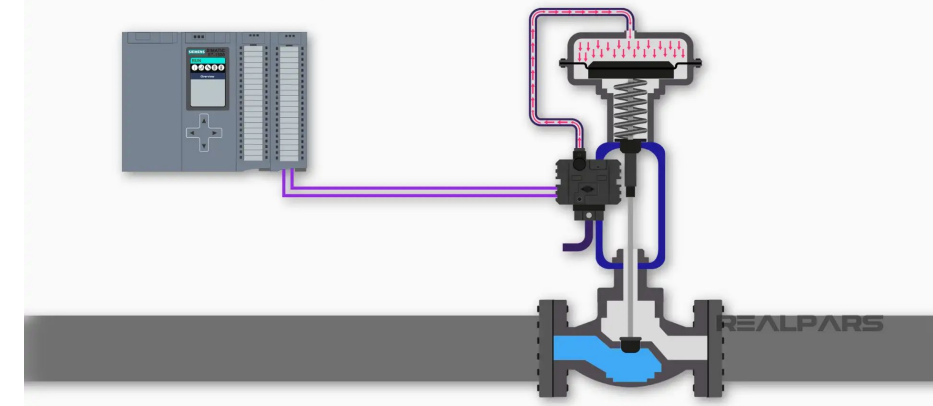
<https://instrumentationtools.com/what-is-a-hydraulic-actuator/>

- Hydraulic power is used in **hydraulic actuators**.
- Hydraulic power - using pressurized fluid to do heavy work with less input effort.
- Pressure of compressed air is used in **pneumatic actuators**



<https://www.realpars.com/blog/pneumatic-actuator>

What is a Pneumatic Actuator?



Examples - Actuators

- Lights, displays, motors, heating or cooling elements, and speakers are some of the examples of **electrical actuators**.
- Another example is a smart home system that uses many sensors and actuators.
- Actuators are used in home applications such as locking and unlocking doors, switching lights on and off, etc.

Electrical Actuator

DC Motor , Servo Motor, Stepper Motor, Linear Actuators, Solenoid, Relay

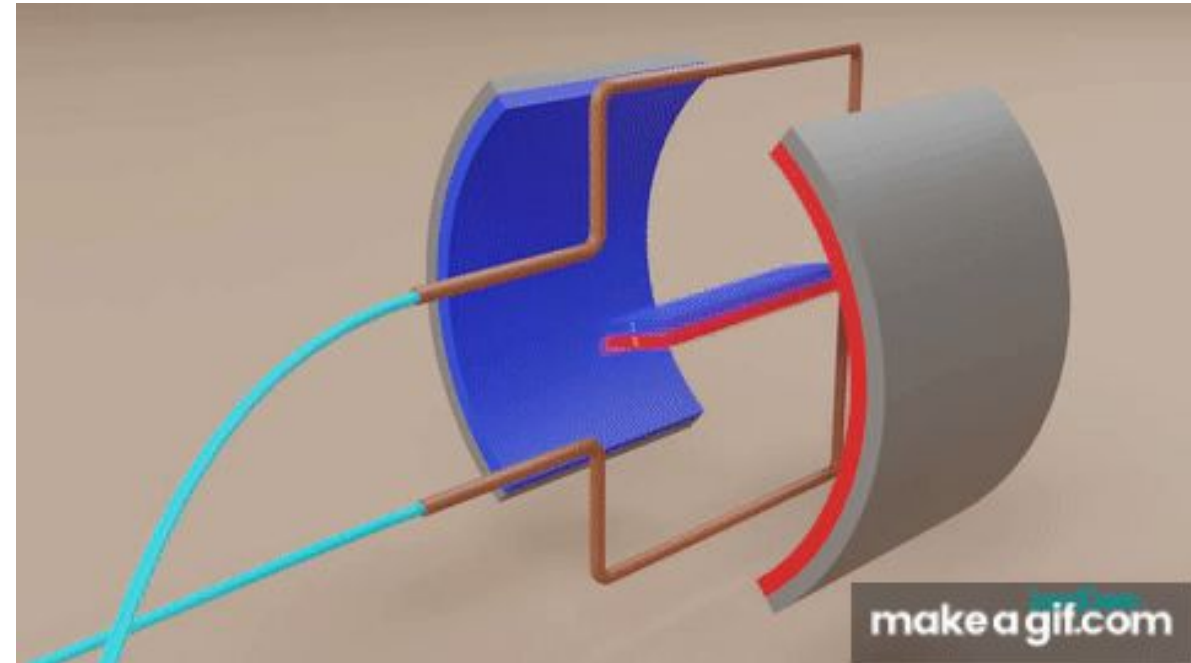
Electric motors

- Electric machines are used for converting energy. Electrical energy is converted by motors to produce mechanical energy.
- Many day-to-day applications use electric motors.
- Two main classes of electric motors: • Direct Current (DC) motor • Alternating Current (AC) motor



DC Motor

- A direct current power motor is known as DC motor.
- Field windings of dc motor are used to provide the magnetic flux and the armature of dc motor acts as the conductor.
- Its working principle is that when a current carrying conductor is placed in a magnetic field, it experiences a force which causes rotation with respect to its original position.
- Dc motor mainly consists of two parts.
 - Rotating part is called the rotor
 - The stationary part is called as the stator.
 - The rotor rotates with respect to the stator



Servo Motor

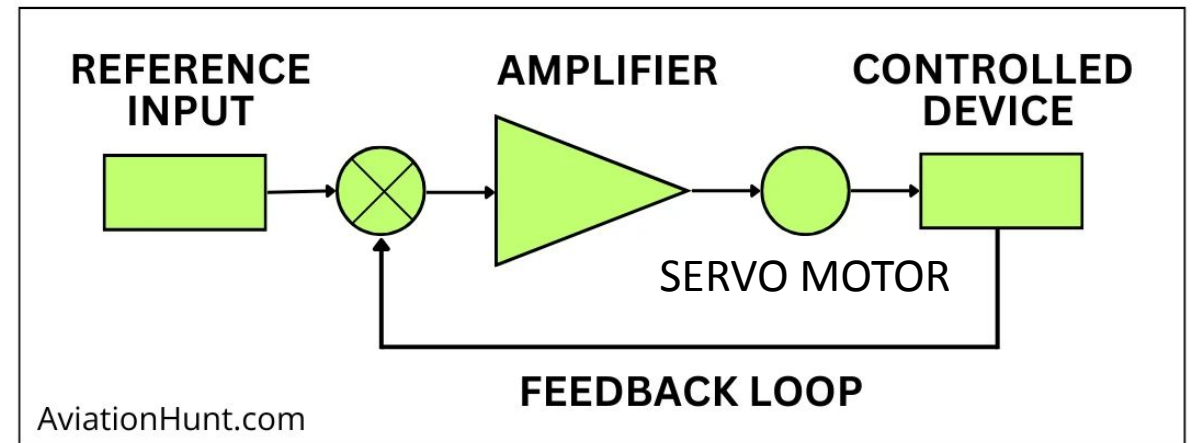
- To push or rotate an object with precision
- an electrical device which can rotate an object at some specific angles or distance.
- It is a simple motor which uses servo mechanism for its operation.
- Operated on DC power □ DC servo motor
- AC powered motor □ AC servo motor.
- Servo motor is available in small and light weight packages with high torque.

Servo Mechanism

- Feedback control system that uses a sensor to monitor its own performance and automatically correct its output to match a desired input, allowing for precise control over a device's position, speed, or torque.

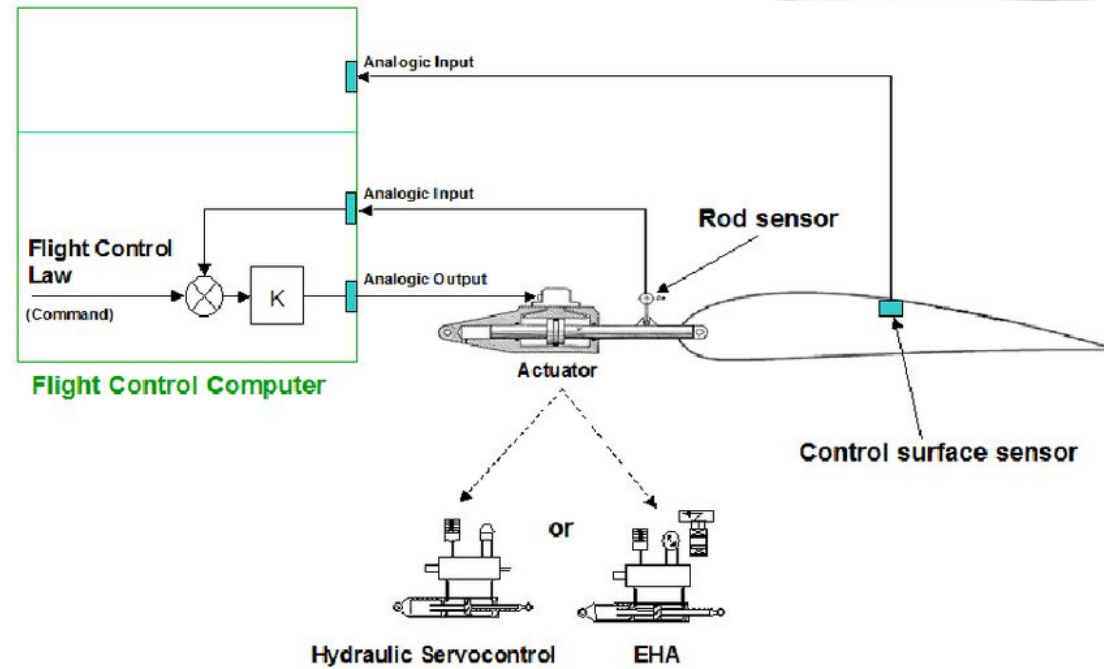


SERVOMECHANISM



Servo Motor - Applications

- Robotics
- RC helicopters
- Planes
- Machine toy cars, etc.



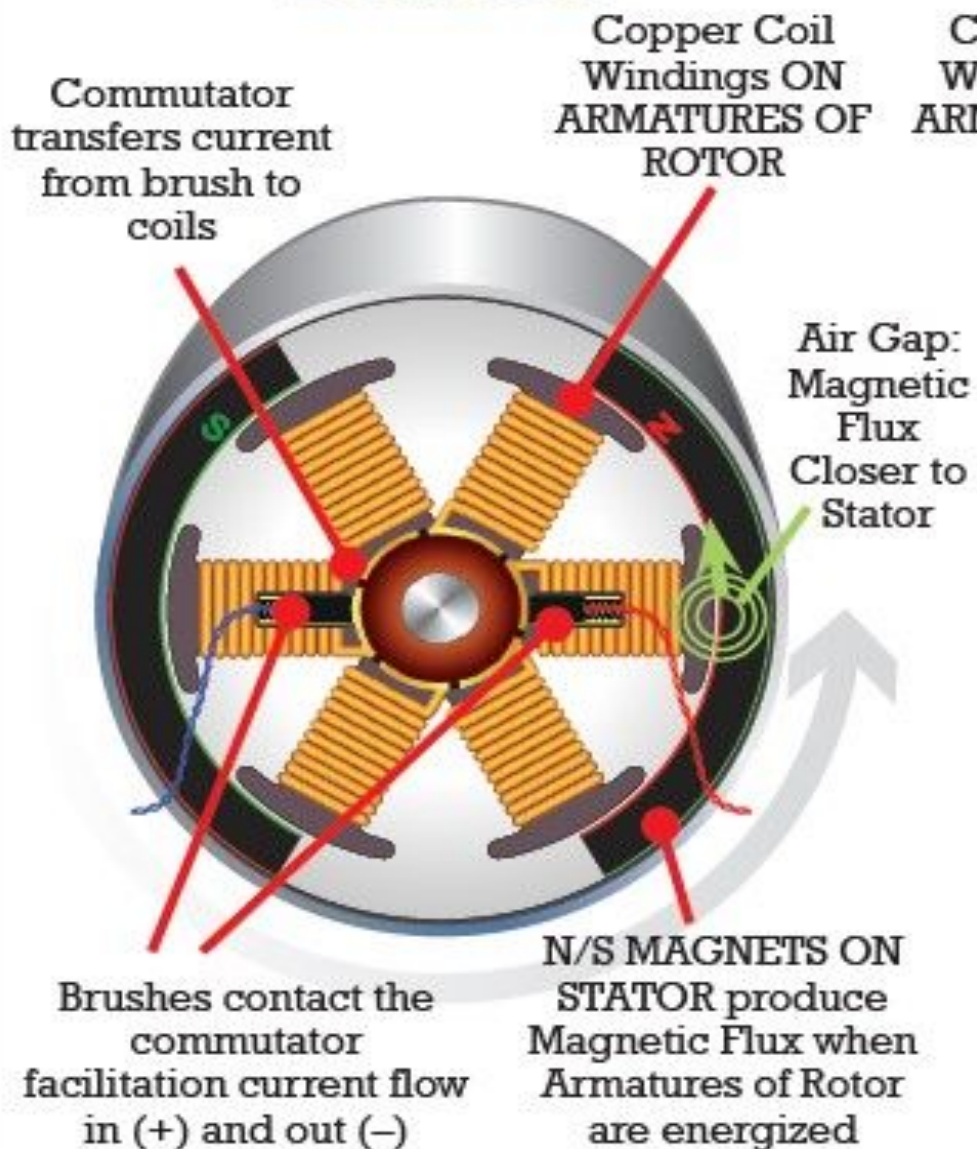
Stepper Motor

- Electromechanical device which converts electrical power into mechanical power.
- It is a brushless, synchronous electric motor and divides its full rotation into an expansive number of steps.
- If the motor is carefully sized to the application, the motor's position can be controlled accurately without any feedback mechanism.

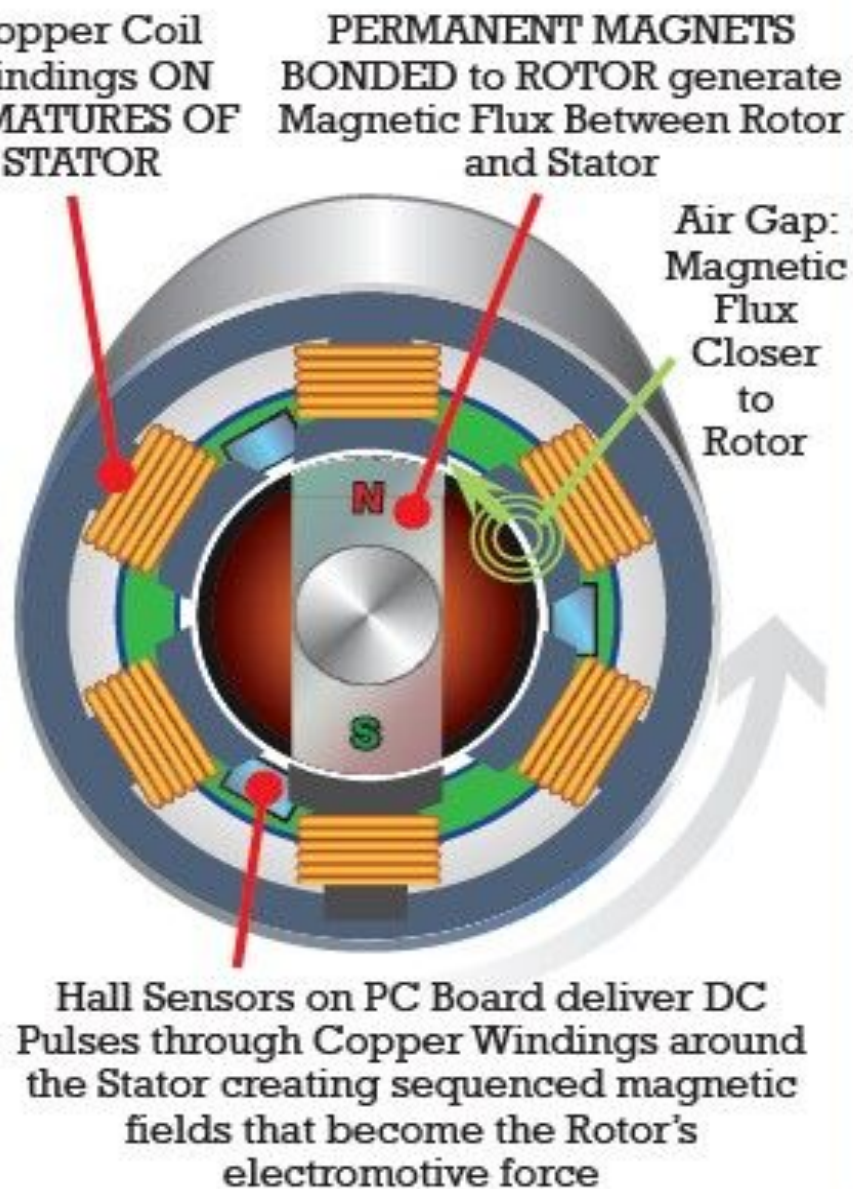
[Brush – links stationary power supply and the rotating coil.

Brushless – Controllers control the power supply to the coil]

Brush Commutated DC Motor



Brushless DC Motor



Stepper Motor Types

Permanent Magnet Stepper Motor (PM Stepper)

- Uses a permanent magnet rotor and a toothed stator.
- Provides good holding torque, suitable for low-speed applications.
- **Use cases:** Robotics, CNC, disk drives, printers (where accuracy matters).

Variable Reluctance Stepper Motor (VR Stepper)

- Has a soft iron (non-magnetized) rotor with salient poles.
- Simple, inexpensive, but lower torque compared to hybrid/PM types.
- **Use cases:** Low-cost automation, toys, instruments (where speed is more important than accuracy).

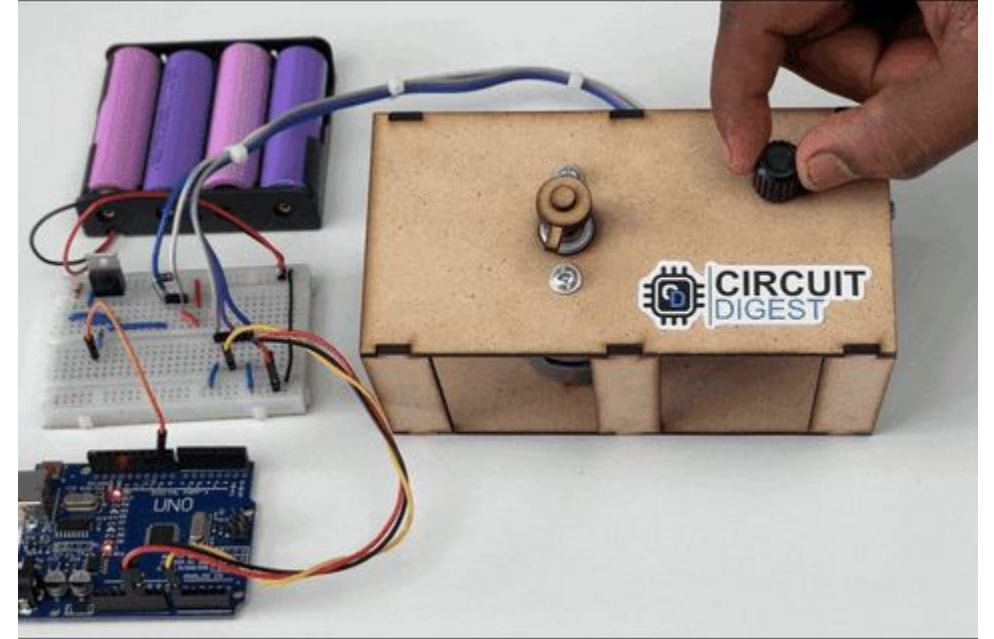
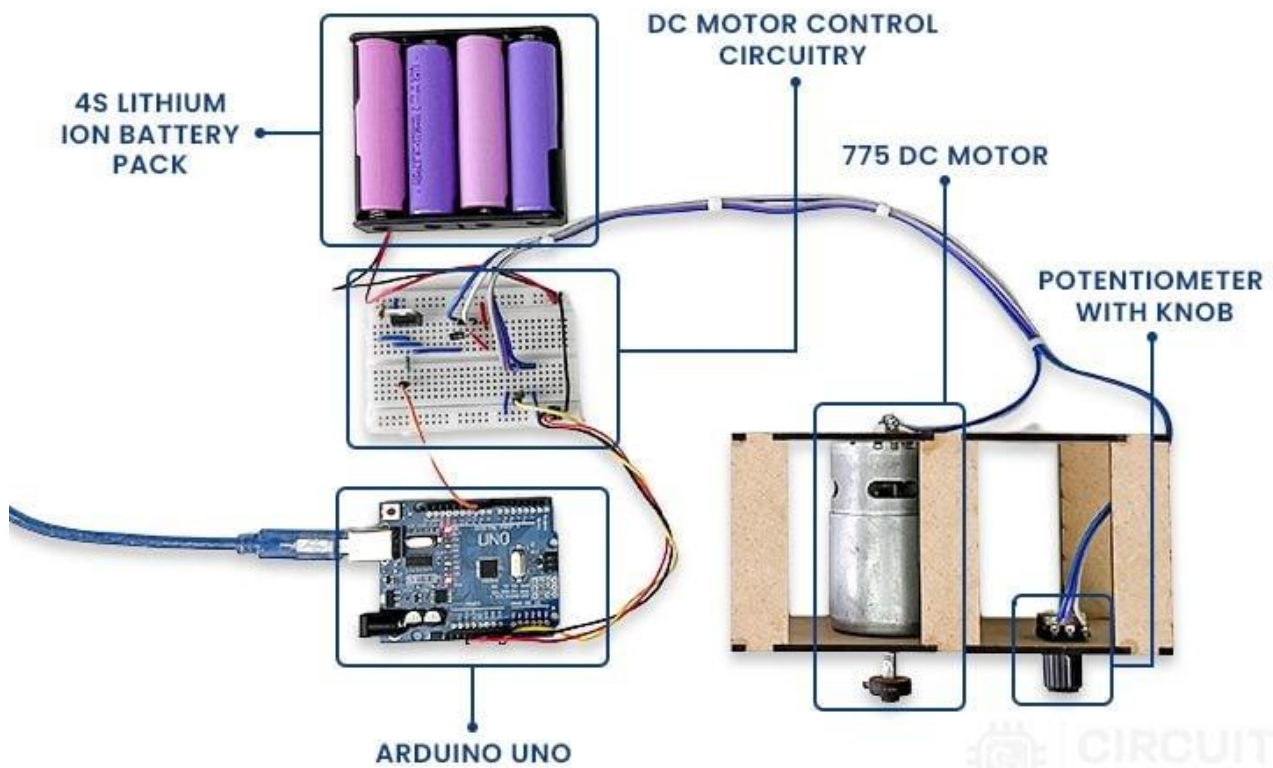
2. Hybrid Synchronous Stepper Motor

- Combines features of permanent magnet and variable reluctance types.
- Offers high torque and better precision, most widely used in industry.

Applications

- Actuators in fluid pumps,
- Medical scanners, samplers, blood analysis machinery, respirators, etc.
- Focus and zoom functions in digital cameras
- Machine tooling equipment, automotive gauges, and surveillance systems for security industry

Motor with Arduino



```
import RPi.GPIO as GPIO
```

```
import time
```

```
# GPIO pins for stepper motor
```

```
pins = [17, 18, 27, 22]
```

```
# Define step sequence  
(half-step mode)
```

```
seq = [
```

```
    [1,0,0,1],
```

```
    [1,0,0,0],
```

```
    [1,1,0,0],
```

```
    [0,1,0,0],
```

```
    [0,1,1,0],
```

```
    [0,0,1,0],
```

```
    [0,0,1,1],
```

```
    [0,0,0,1]
```

```
]
```

```
GPIO.setmode(GPIO.BCM)
```

```
for pin in pins:
```

```
    GPIO.setup(pin, GPIO.OUT)
```

```
    GPIO.output(pin, 0)
```

```
def step_motor(steps, delay=0.01):
```

```
    for _ in range(steps):
```

```
        for pattern in seq:
```

```
            for pin, val in zip(pins, pattern):
```

```
                GPIO.output(pin, val)
```

```
            time.sleep(delay)
```

```
try:
```

```
    print("Stepper motor ON: Rotating 1 revolution")
```

```
    step_motor(512) # 512 steps  $\approx$  1 revolution for 28BYJ-48
```

```
    print("Motor OFF")
```

```
    for pin in pins:
```

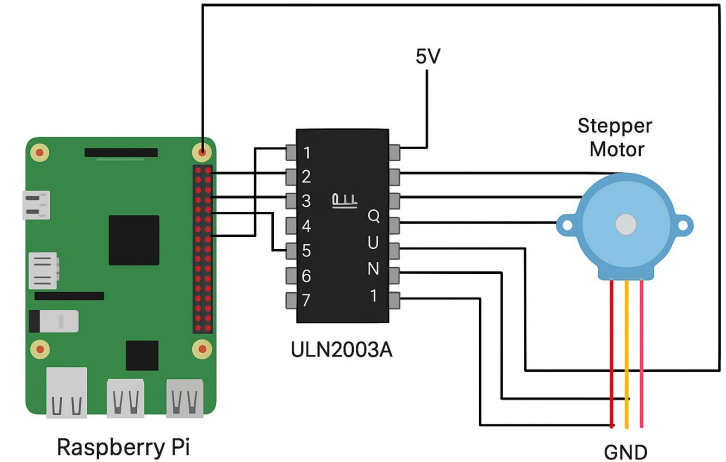
```
        GPIO.output(pin, 0)
```

```
except KeyboardInterrupt:
```

```
    pass
```

```
finally:
```

```
    GPIO.cleanup()
```



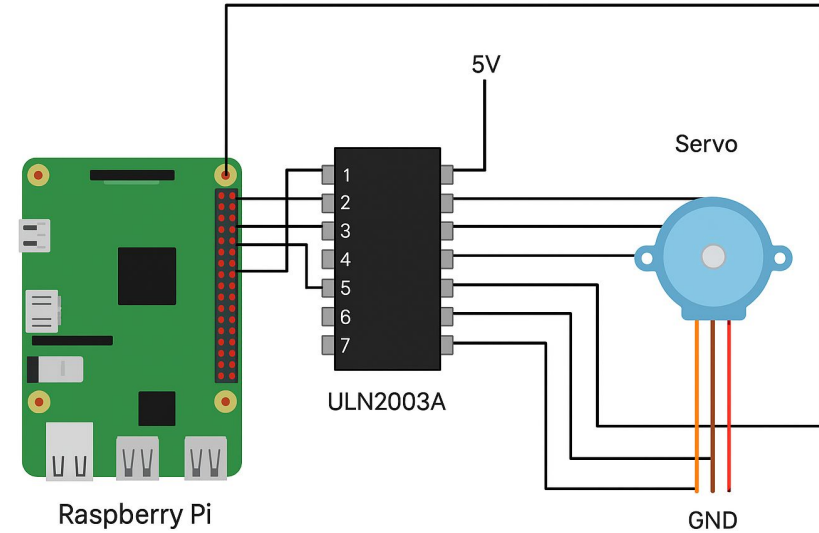
```
import RPi.GPIO as GPIO
import time
```

```
GPIO.setmode(GPIO.BCM)
GPIO.setup(18, GPIO.OUT)
```

```
servo = GPIO.PWM(18, 50) # 50 Hz
servo.start(0)
```

```
print("Servo ON: moving")
servo.ChangeDutyCycle(7) # mid position
time.sleep(1)
servo.ChangeDutyCycle(12) # rotate one side
time.sleep(1)
servo.ChangeDutyCycle(2) # rotate other side
time.sleep(1)
```

```
print("Servo OFF")
servo.stop()
GPIO.cleanup()
```



Linear Actuators

- The type of actuator is useful when motion in a straight line (linear motion) between two points is required.
- These could be mechanical or electro-mechanical (also known as electric linear actuators) devices.
- The rotation of the electric motor is converted to linear motion by using gears and lead screw in the electric linear actuators.

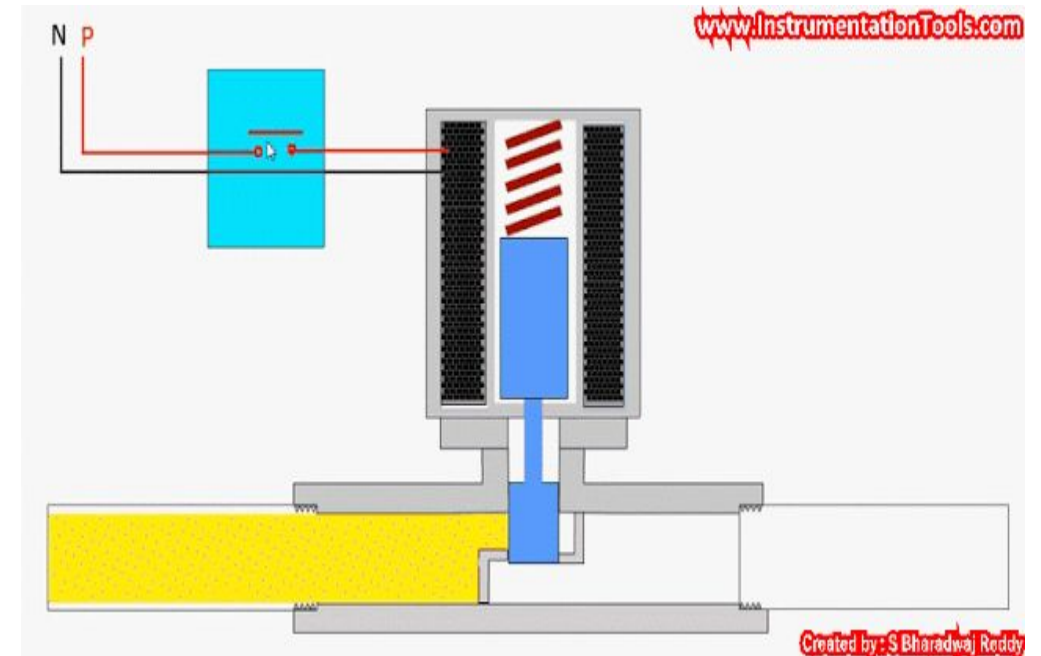


Linear Actuators - Applications

- Valves
- Automatic windows/doors
- Agriculture implements
- Robotics
- Machine design
- Material management
- Industrial equipment, etc.

Solenoid

- A coiled wire around a cylinder is called as a solenoid.
- When current runs through a solenoid, a strong magnet is generated because the magnetic field is concentrated inside the coil.
- Unlike regular magnets, an electromagnet is particularly useful as it can be switched on and off, and its strength increases by increasing the amount of current flowing through it.
- This phenomenon is useful in many applications.

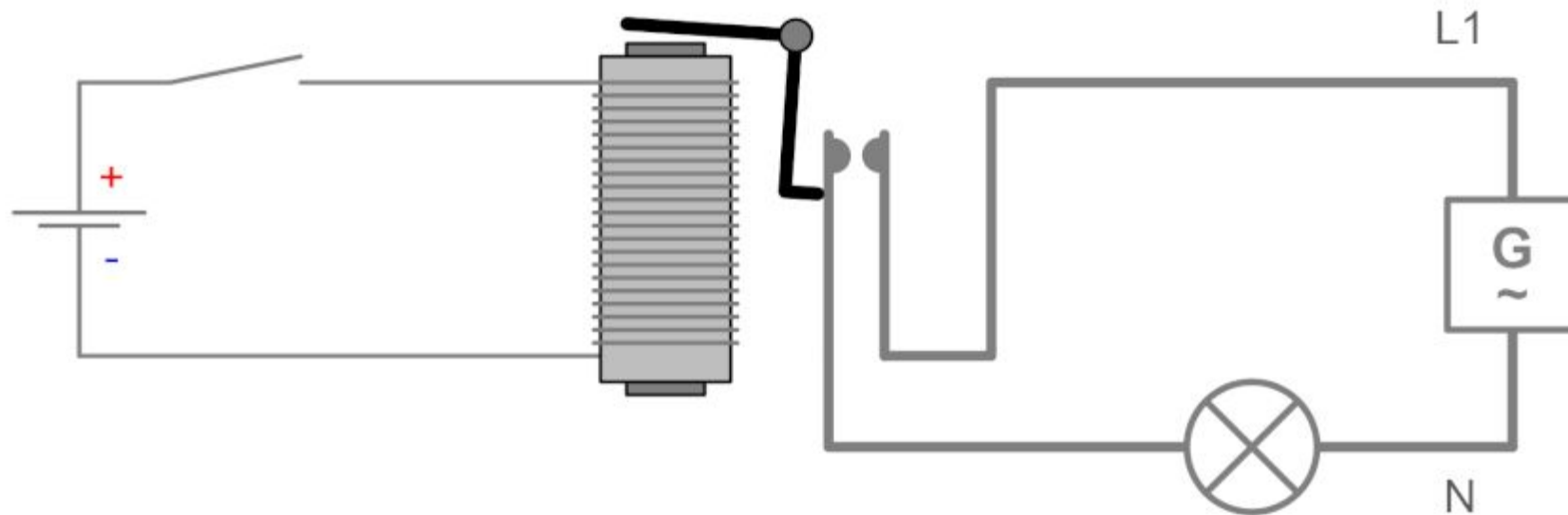


Electromagnetic Solenoid and Its Use

- Valves (air compressors, metres),
- Pistons,
- Home automation devices (door bells, auto locks, AC, heating equipment),
- Automobiles (engine starter),
- Drinks, coffee dispensers, sprinkler system, etc.

Relay

- The relay device receives input signal from one side and controls the switching operation on the other side.



Relays

- Many types of relays are present.
- A commonly used relay is made of electromagnets and operates as a switch.
- Relay is a switch that controls the circuit electromechanically.
- A high-powered circuit is controlled by a relay switch with a low power signal.
- Generally, a DC signal is used to control a circuit that drives high voltage appliances such as smart home appliances that operate on AC power supply

Different Mechanical Parts of a Relay

- Electromagnet
- movable armature
- contacts
- spring (optional)
- yoke

When the relay is on, the armature moves and connects the normally opened contact pin of the relay. The circuit is closed and the current flows through the coil and comes back to the original position after it is de-energized.

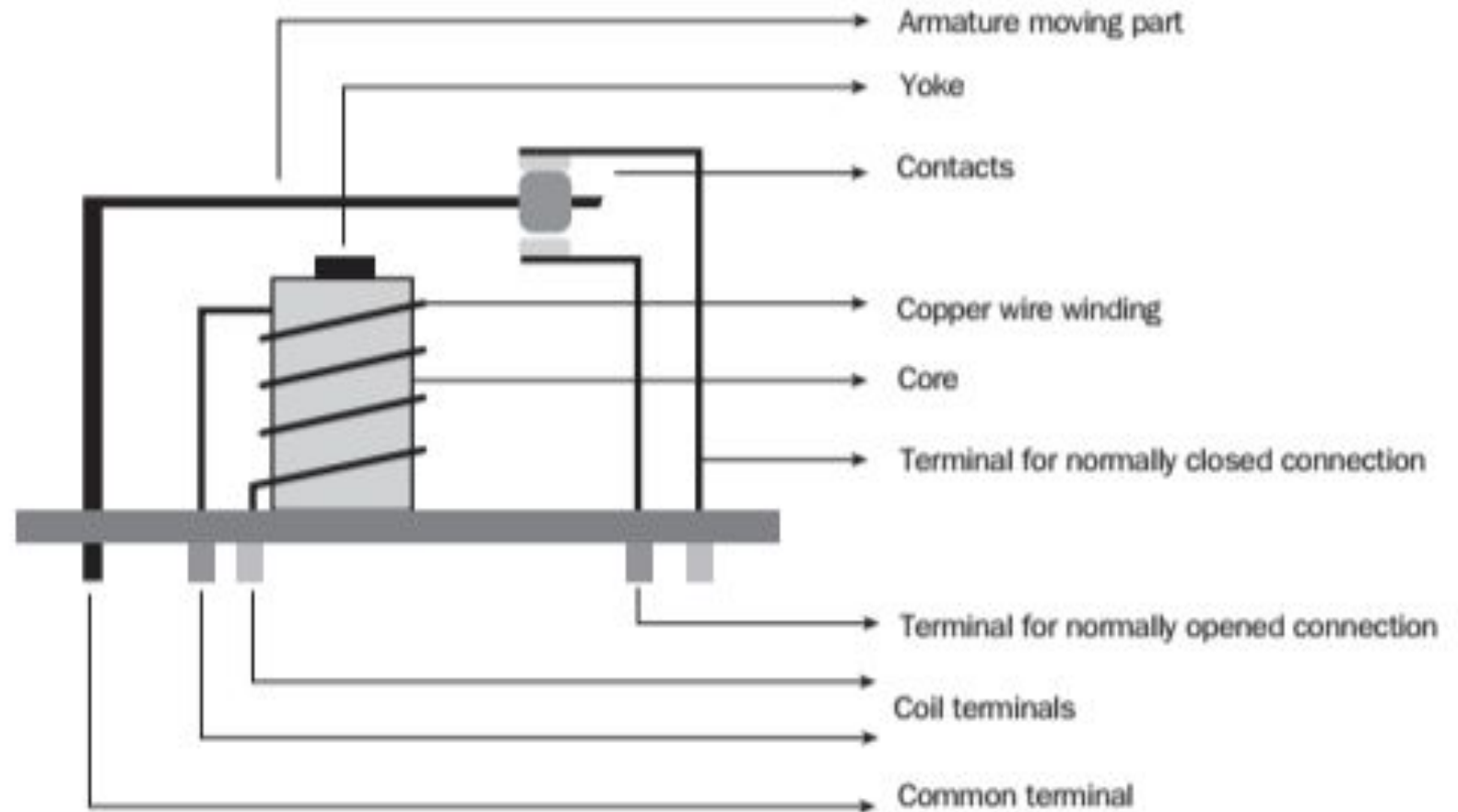


Fig. 5.11 Electromagnetic Relay

Circuit Representation of a relay

NC: Normally closed

NO: Normally opened

SPDT relay: Single Pole Double Throw relay

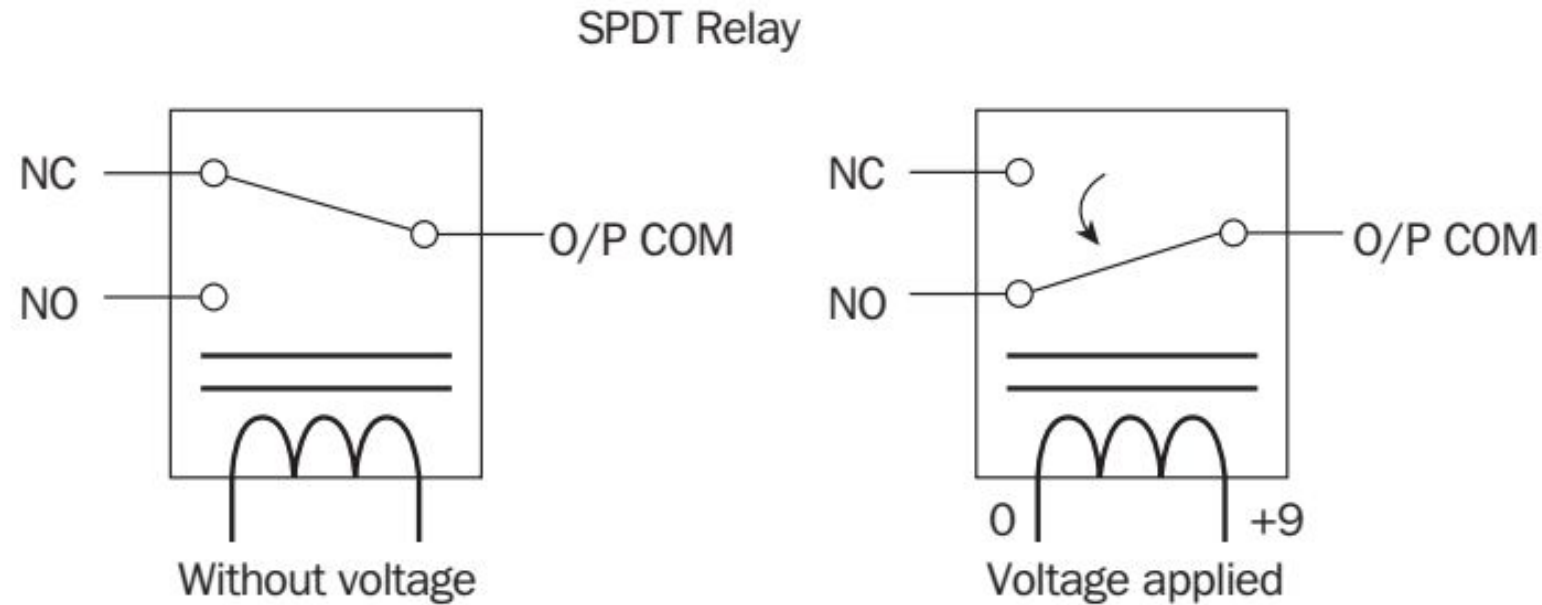


Fig. 5.12 Circuit of a Relay

Different Types of Relay

- **Based on the Principle of Operation**

- 1. Solid State Relay (SSR)**

- Uses semiconductor devices (like transistors, TRIACs).
 - No moving parts, faster switching, long life.

- 2. Electrothermal Relay**

- Works on thermal expansion of metals (bimetallic strip) due to current.
 - Commonly used as overload protection in motors.

- 3. Electromechanical Relay (EMR)**

- Uses electromagnet + mechanical contacts.
 - Widely used, simple, but slower and subject to wear.

- 4. Hybrid Relay**

- Combines solid-state and electromechanical features.
 - Solid-state for fast switching, mechanical contacts for low on-state loss.

- **Based on the Polarity**

- 1. Polarized Relay**

- Coil has a **defined polarity**, needs correct current direction.
 - More **sensitive and precise** operation.

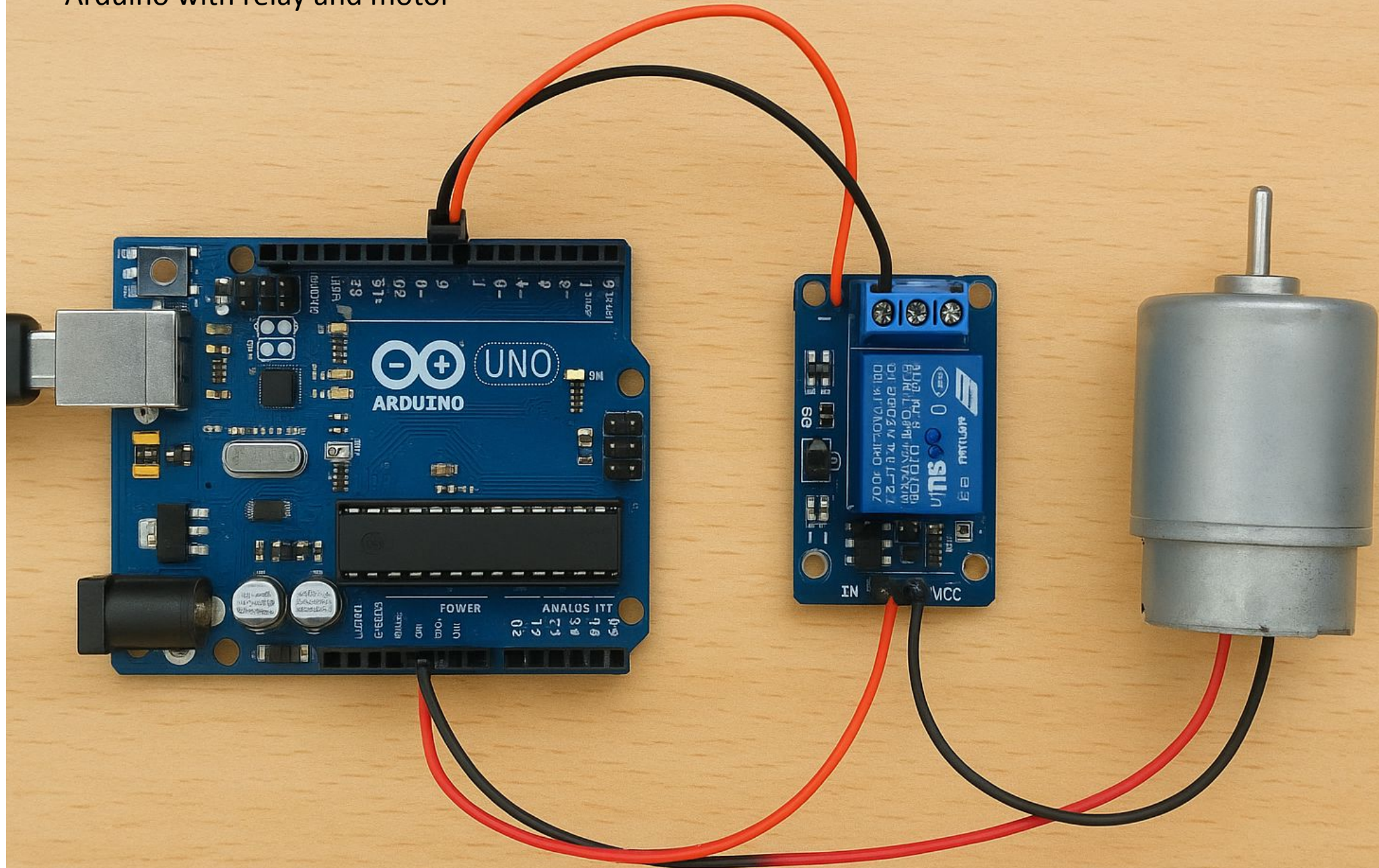
- 2. Non-polarized Relay**

- Coil works with **any current direction**.
 - **Simpler and robust**, less sensitive.

Applications of Relay

- Applications of relay include
 - Home appliances,
 - Home/office automation (temperature-based control of heating/cooling, lighting),
 - Traffic signals control,
 - Industrial applications,
 - Precision agriculture,
 - Automotive applications, etc.

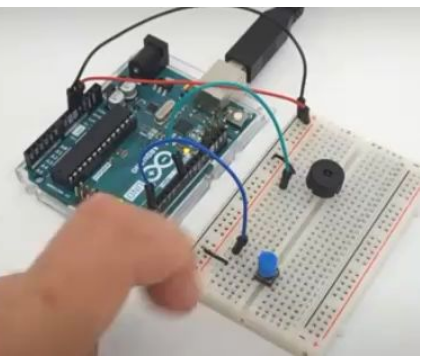
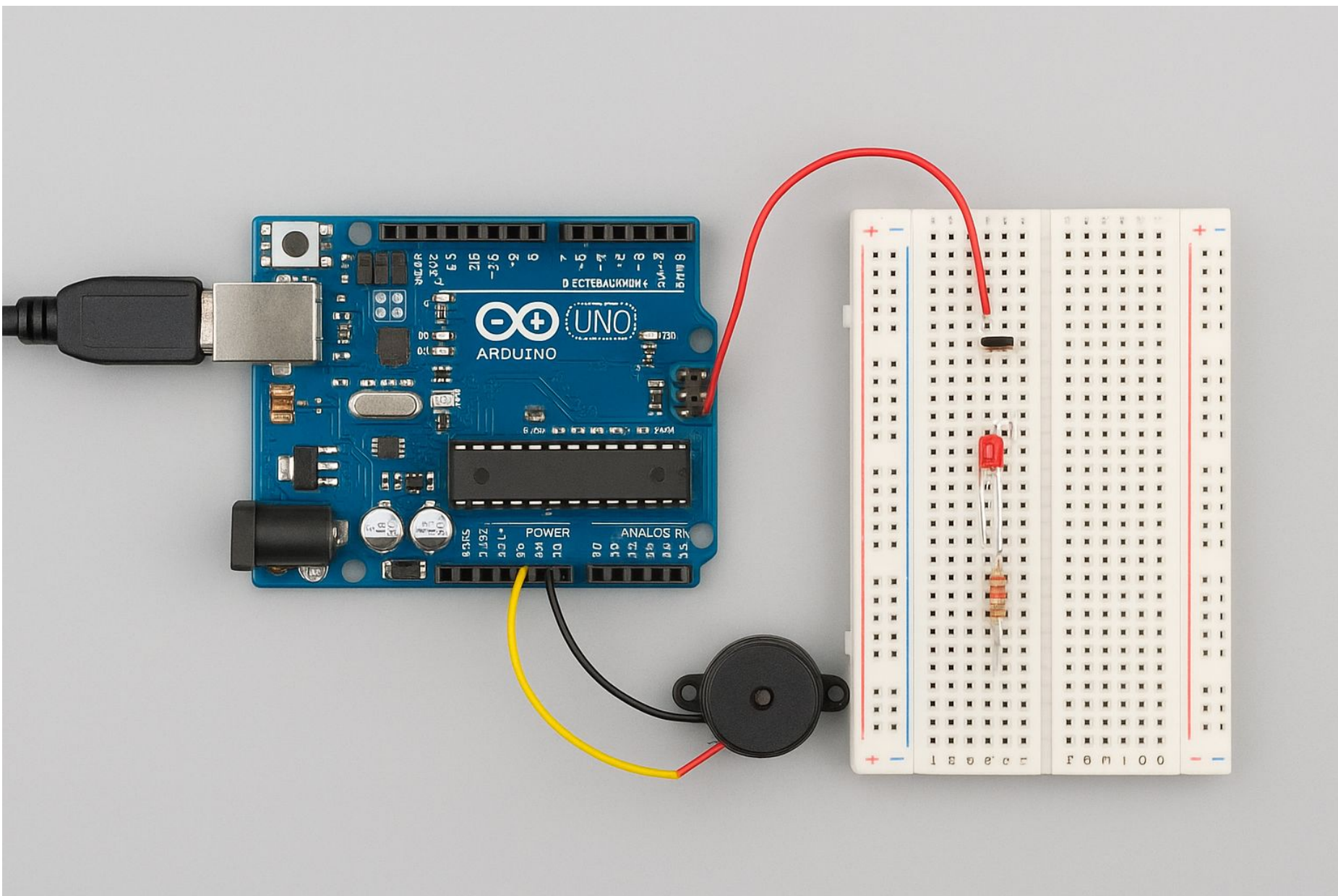
Arduino with relay and motor



IoT Example

Light Sensitive Alarm System

- Design and implement an Arduino-based light monitoring and alarm system using an **LDR (Light Dependent Resistor)**, **LED**, and **buzzer**. The system should continuously read the light intensity from the LDR connected to analog pin **A0** and display the values on the Serial Monitor.
- If the **LDR reading is greater than 100**:
 - The LED (pin **12**) should blink.
 - The buzzer (pin **13**) should beep.
- If the **LDR reading is less than or equal to 100**:
 - The LED should remain OFF.
 - The buzzer should remain OFF.
- This simulates a **light-sensitive alarm system**, where the alarm (LED + buzzer) is activated under a certain light threshold.



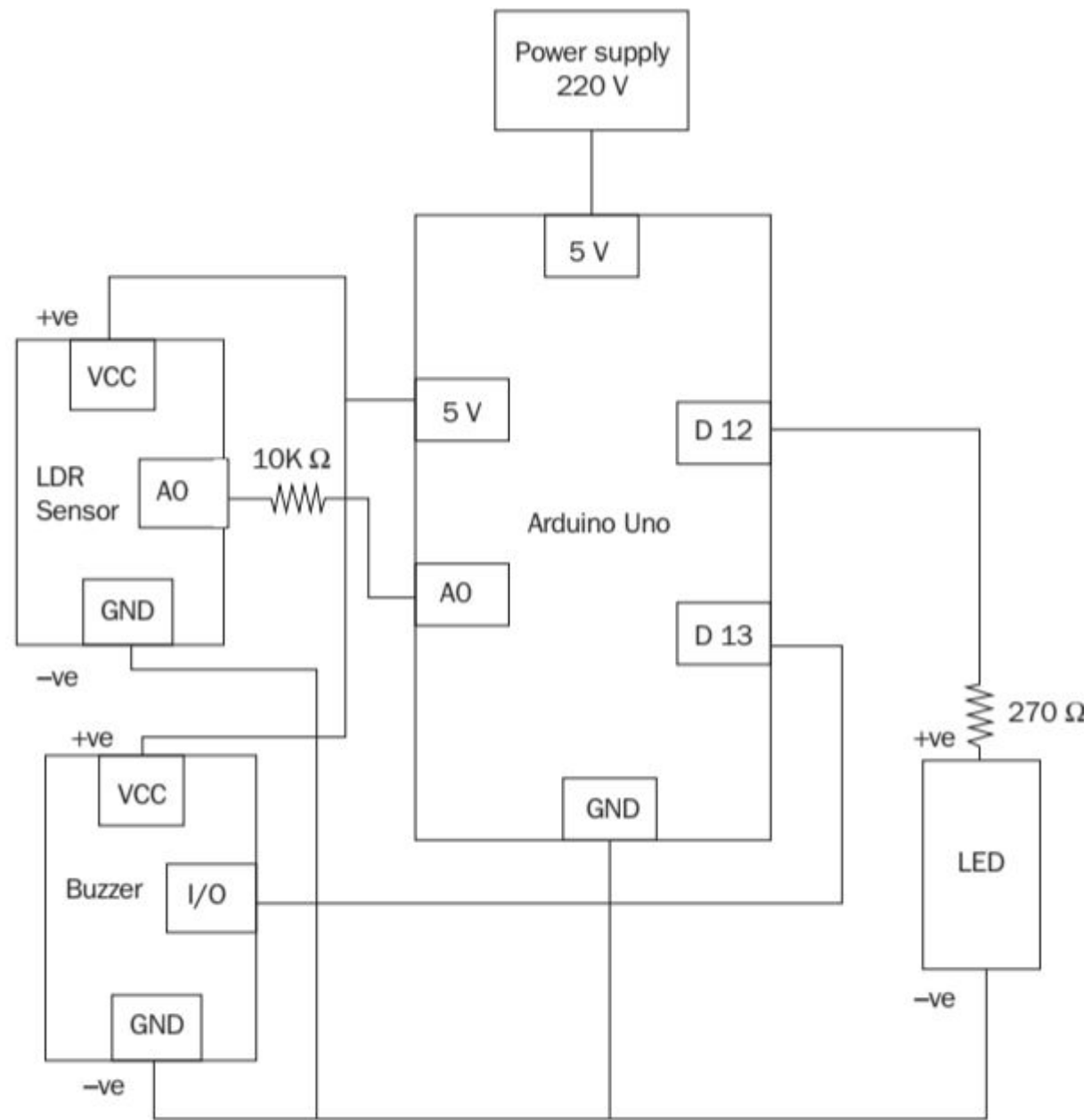


Fig. 5.13 Circuit Diagram for Sensor as Light Dependent Resistor (LDR) and Actuators as LED and Buzzer

LDR Configuration

```
const int ldr_pin = 7;
const int led_pin = 13;

void setup() {
    pinMode(ldr_pin, INPUT);
    pinMode(led_pin, OUTPUT);
    Serial.begin(9600);
}

void loop() {
    if ( digitalRead( ldr_pin ) == 1 ){
        digitalWrite( led_pin, HIGH);
    }
    else {
        digitalWrite( led_pin , LOW);
    }
    Serial.println( digitalRead( ldr_pin ));
    delay(100);
}
```

Buzzer Configuration

```
const int buzzer = 9;

void setup() {
    pinMode(buzzer, OUTPUT);
    Serial.begin(9600);
}

void loop() {
    tone(buzzer, 1000, 500); // Play 1kHz sound for 500ms
    Serial.println("Buzzer beeps!");
    delay(1000);

    noTone(buzzer); // Stop the sound
    Serial.println("Buzzer do not beep!");
    delay(1000);
}
```

Pseudocode

- In the function void setup(), we initialize analog pin A0, as the input, pin 12 and 13 as the output
- We read the input A0, and print it on the serial monitor
- If the ldrStatus is greater than 100. LED blinks and the buzzer beeps or the alarm is deactivated and our LED stops blinkin

```
int ldrPin = A0;    // LDR connected to analog pin A0
int ledPin = 12;    // LED connected to digital pin 12
int buzzerPin = 13; // Buzzer connected to digital pin 13

int ldrStatus = 0;

void setup() {
  pinMode(ldrPin, INPUT);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
  Serial.begin(9600);    // Start serial monitor
}
```

```
void loop() {
  ldrStatus = analogRead(ldrPin);    // Read LDR value
  Serial.println(ldrStatus);          // Print to serial monitor

  if (ldrStatus > 100) {
    // LED blinks
    digitalWrite(ledPin, HIGH);
    tone(buzzerPin, 1000, 500);    // Play 1 kHz sound for 500 ms
    delay(500);

    digitalWrite(ledPin, LOW);
    noTone(buzzerPin);              // Stop sound
    delay(500);
  } else {
    // LED and buzzer off
    digitalWrite(ledPin, LOW);
    noTone(buzzerPin);
  }
}
```

Exercises

- Design a sensor and actuating circuit for temperature and humidity sensor.
- Write a code in Python or Embedded C for collecting the information from a temperature and humidity sensor.
- Write a Python code for generating an output for actuating a fan using a relay.